Documentation for Anybus Modbus-TCP/RTU Configuration File for °CaleonBox-GLT

This documentation is presenting the different options for using the GLT-functions of the °CaleonBox. The reading and writing registers are set with the „Reassign Addresses“-tool, so they are put in the same order as in the configuration tool. This order can be changed manually.

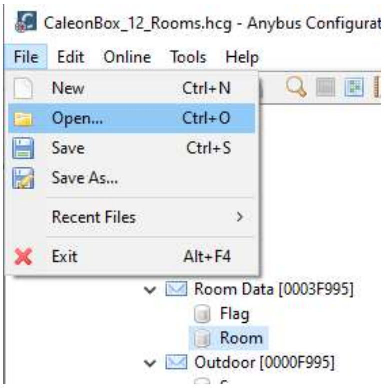
A configuration with all objects beneath and twelve rooms is already designed and can be used as a base for future configurations. For more information contact the Sorel GmbH Support.

1. Anybus Communicator:

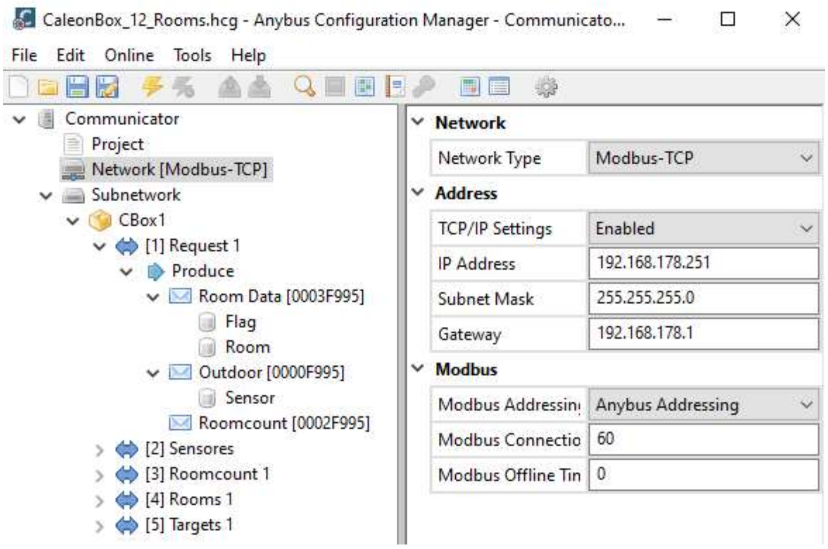
a. Please check user manual Anybus Modbus TCP or Anybus Modbus RTU
b. Connect the device with the included USB-Cable to your PC
c. Install the configuration tool Anybus Configuration Manager

2. Start the Anybus Configuration Manager

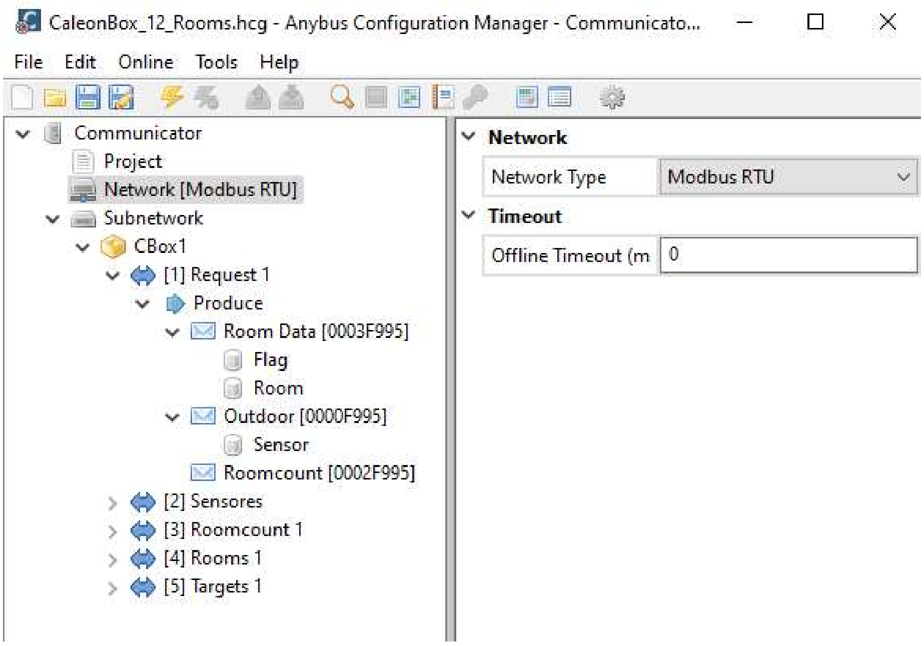
a. Open an existing file or create a new one (*.hcg)



b. For Modbus-TCP select an IP-address or use the stardard address



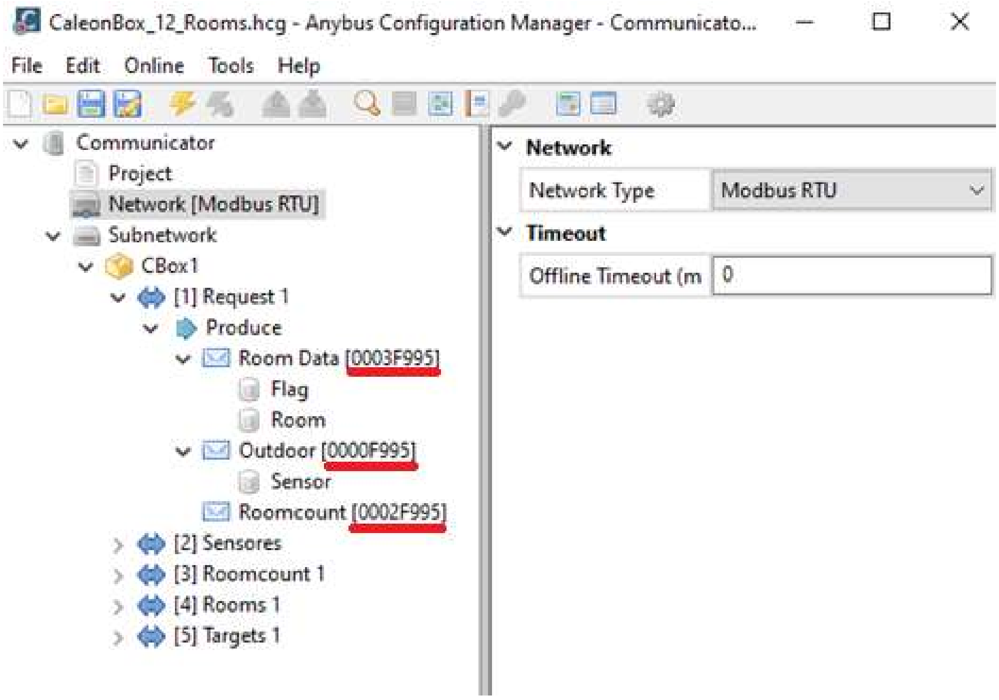
c. For Modbus-RTU just change the network type



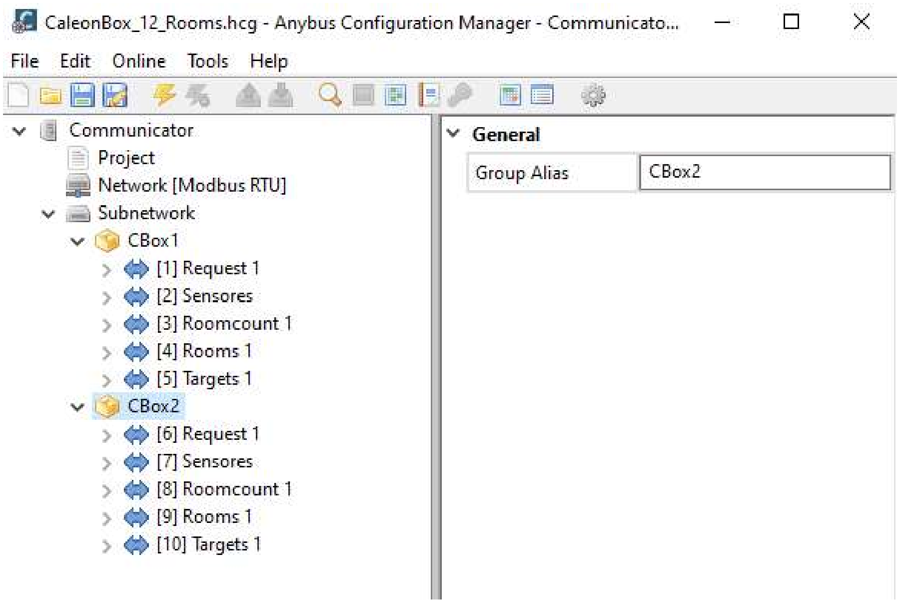
Important: When using Modbus RTU make sure to use the right USB – RS232-Cable. Also check if the baud rate, the node ID, parity and data bits are the same (Communicator, COM-Port and Modbus-Master e.g. Modbus Poll)

d. The CAN-Frames are different for each controller. Each CAN-Frame contains the CAN-ID of the used device.

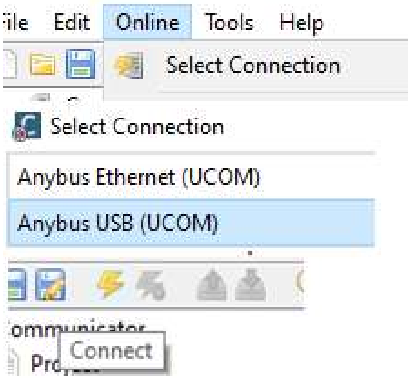
°CalenBox Room Data Request (0x0003<CAN-ID>95)
The CAN-ID has to be written in HEX format and hast o be set for each used CAN-Frame
Example: CAN-ID of the controller is ***249*** *which is* ***F9*** *written in HEX format ->* ***0x0003F995***



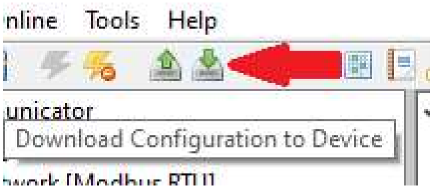
e. Repeat for each station



f. Connect to the Anybus Communicator



g. Download the configuration file to the device



Anybus Konfiguration for °CaleonBox

This documentation should help setting up an Anybus CAN-Converter with a °CaleonBox. This is only an addition to the CAN-Documentation. For more information and information about the implementation please check the CAN-Documentation.

1. Requesting Information

a. Select 29-bit CAN Address in Subnetwork
b. Add Produce
c. Rename the Transaction
d. **Room data:** CAN Frame [0003XX95]
   i. Flag 2 Byte (Temperature=0x1, Humidity=0x2, Location=0x?, Tset=0x10, Hysteresis=0x20)
   ii. Room 2 Byte (Room-ID)
e. **Sensore:** CAN Frame [0000XX95]
   i. Sensor 2 Byte (Outdoor=0x0)
f. **Roomcount:** CAN Frame [0002XX95]
   i. No data, No register

- Produces are mapped to 0x200 (can be check at „Address Overview)
- Anybus is reserving the registers 256-1023 for some reason
- Produces start at register 1024

2. Reading Information

a. Select 29-bit CAN Address in Subnetwork
b. Add Consume
c. Rename the Transaction
d. **Sensore:** CAN Frame [1000XX95]
    i. Sensor 4 Byte
    ii. Select Word Swap for reading „Signed“-formate
e. **Roomcount**: CAN Frame [1002XX95]
    i. Count 2 Byte
    ii. Select Word Swap for reading „Signed“-formate
f. **Room data:** CAN Frame [1003XX95]
    i. CAN Identifier (DWord) for selecting temperature or humidity of a certain room
    ii. Examples:

```
Temperature Room 1 -> CAN Identifier (Dword): 0x00001

Humidity Room 1 -> CAN Identifier (Dword): 0x00002

Temperature Room 2 -> CAN Identifier (Dword): 0x10001

Humidity Room 2 -> CAN Identifier (Dword): 0x10002

…
```

    iii. Data 2 Byte
    iv. Select Word Swap for reading „Signed“-formate

3. Changing Target temperatures

a. Select 29-bit CAN Address in Subnetwork
b. Add Produce
c. Rename the Transaction
d. Use the same Adress ID as reading room data
    i. CAN Identifier (DWord) for selecting TSet or Hysteresis of a certain room
    ii. Examples:

```
TSet Room 1 -> CAN Identifier (Dword): 0x00010

Hysteresis Room 1 -> CAN Identifier (Dword): 0x00020

TSet Room 2 -> CAN Identifier (Dword): 0x10010

Hysteresis Room 2 -> CAN Identifier (Dword): 0x10020

…
```

    iii. Data 2 Byte
    iv. Select Word Swap for reading „Signed“-formate

4. Additional Information

a. By selecting „Reassign Addresses“ the mapping is automatically starts at register 0 for RESPONSES and register 1024 for REQUESTS
b. „XX“ in CAN Frames needs to be replaces by the CAN-ID as HEX
c. The request update time can be changed for every produce. Requests can also be triggered by changing the value oft he registers (needs to be set in produce settings)

d. IP Address can be changed in Network [Modbus-TCP], if TCP/IP Settings are „Enabled“

i. Reqeust Value in room flag register for temperature: 0x0001

Example for Room 1:

	Alias	01024
1024	Roomflag	(??) 0x0001
1025	Room	(??) 0x0000

ii. Request Value in room flage register for humidity: 0x0002

Example for Room 1:

	Alias	01024
1024	Roomflag	(??) 0x0002
1025	Room	(??) 0x0000

iii. Request Value for room: 0x000<room number – 1>

Example for humidity of Room 4:

	Alias	01024
1024	Roomflag	(??) 0x0002
1025	Room	(??) 0x0003